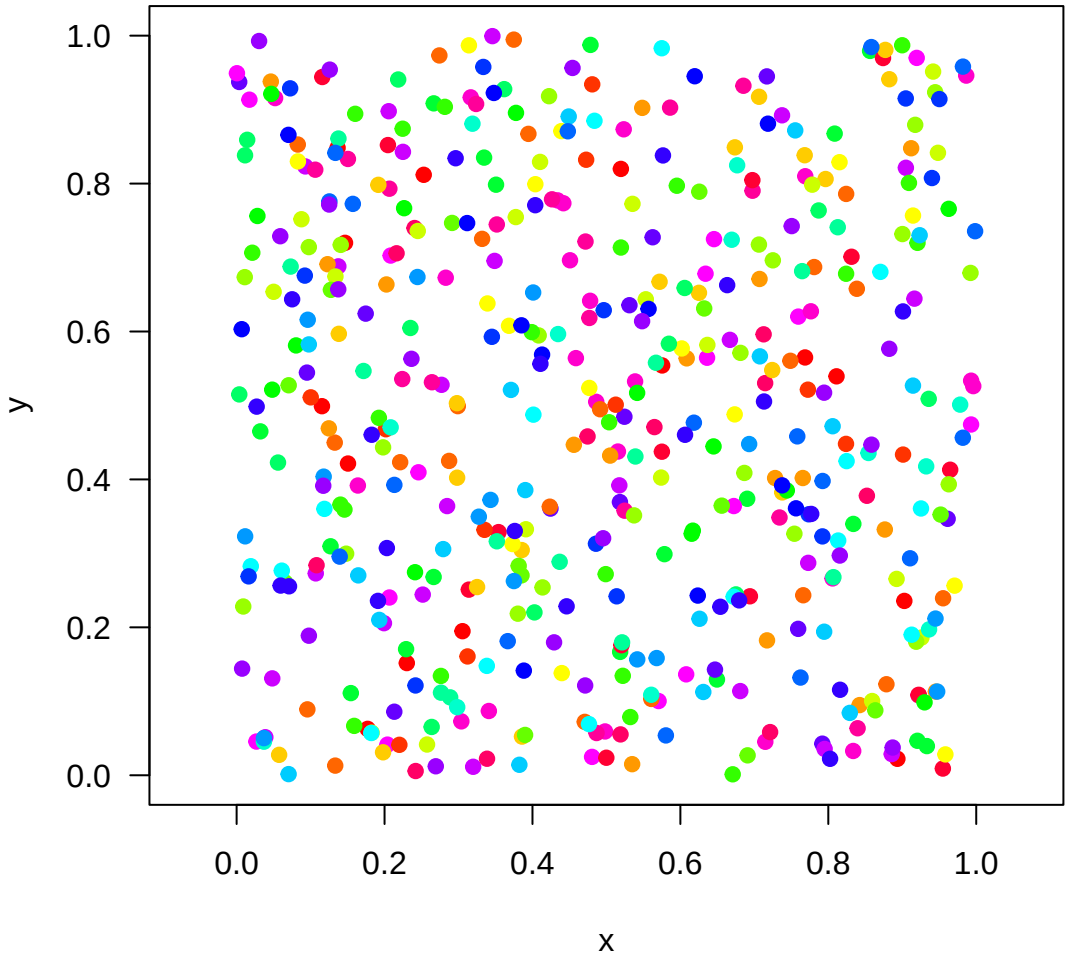
Simulated communities & sampling for the jelly excersize

25 January 2023

```
#install.packages("mobsim")
#install.packages("iNEXT")
library(mobsim)
library(vegan)
library(tidyverse)
library(iNEXT)
```

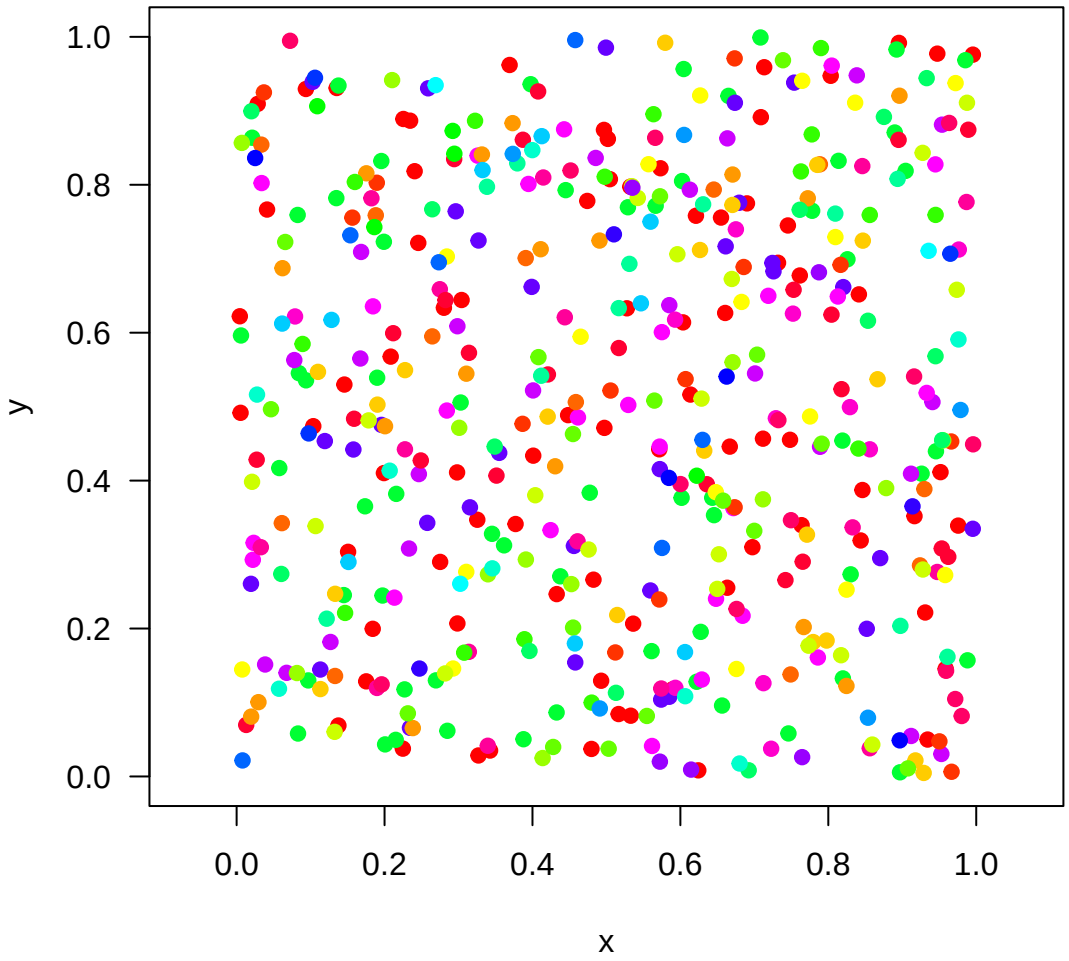
Simulate an even and an uneven community with random spatial positions

```
# Even community
set.seed(42)
com1 <- sim_poisson_community(s_pool = 30, n_sim = 500, sad_type = "lnorm",
                              sad_coef = list("sdlog" = 0))
plot(com1)
```



```r
# Uneven community
com2 <- sim_poisson_community(s_pool = 30, n_sim = 500, sad_type = "lnorm",
                              sad_coef = list("sdlog" = 1))

plot(com2)
```




```r
# Real diversity

# Community 1

count1 <- com1$census %>%
  group_by(species) %>%
  summarize(count = as.numeric(length(species))) %>%
  column_to_rownames(var = "species")

count.t <- t(count1)

# Diversity
exp(diversity(count.t, "shannon"))
```

```
## [1] 29.29194
```

```
(diversity(count.t, "invsimpson"))
```

```
## [1] 28.61067
```

```
# Community 2

count2 <- com2$census %>%
  group_by(species) %>%
  summarize(count = as.numeric(length(species))) %>%
  column_to_rownames(var = "species")

count2.t <- t(count2)

# Diversity
exp(diversity(count2.t, "shannon"))
```

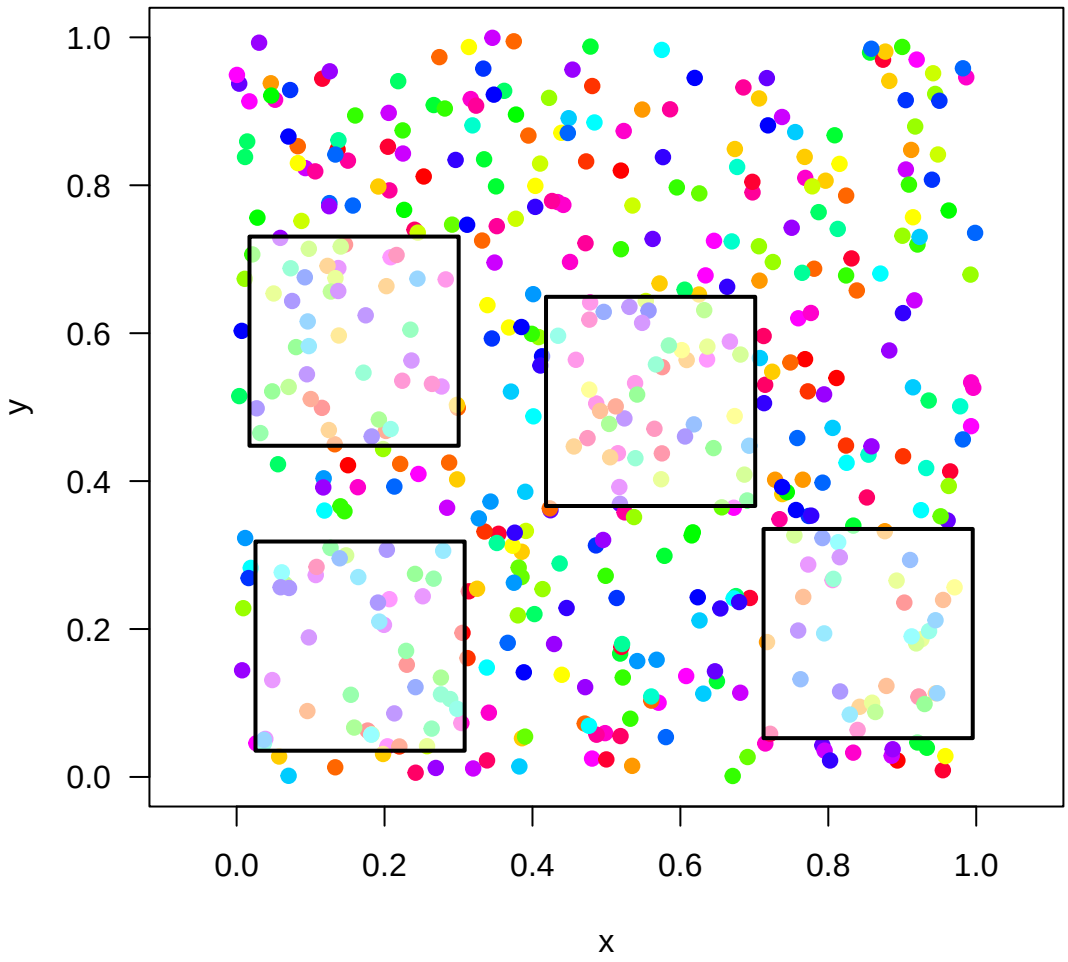
```
## [1] 21.54639
```

```
(diversity(count2.t, "invsimpson"))
```

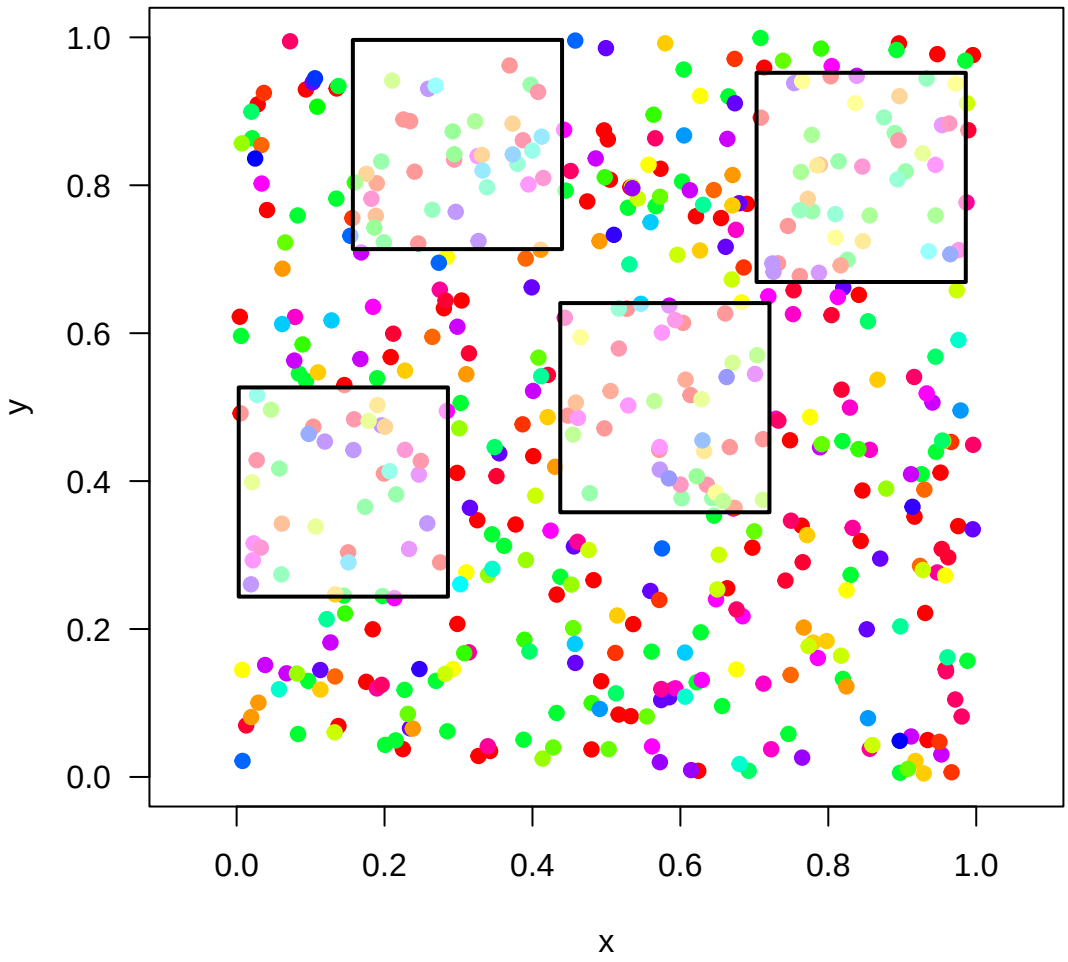
```
## [1] 15.73317
```

Take samples from source communities spatially-explicit census

```
comm_mat1    <- sample_quadrats(com1, n_quadrats = 4,
quadrat_area = 0.08, method = "random", avoid_overlap = T)
```




```
comm_mat2    <- sample_quadrats(com2, n_quadrats = 4,
quadrat_area = 0.08, method = "random", avoid_overlap = T)
```



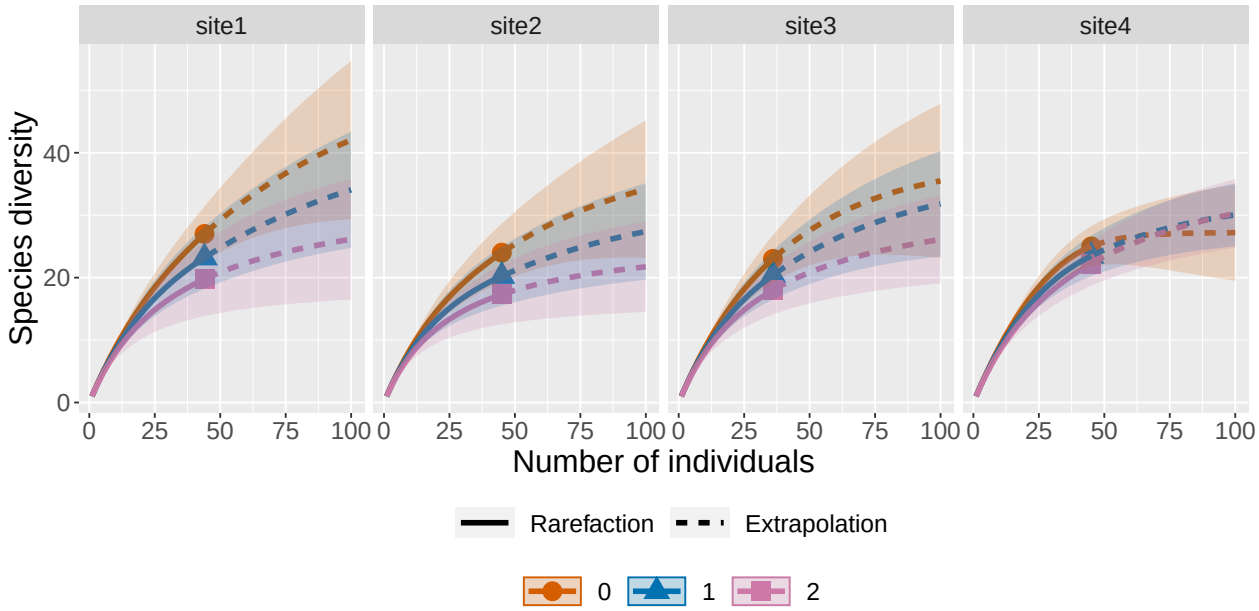
Graphical exploration

```
data1       <- t(comm_mat1$spec_dat)
data1.clean <- data1[, colSums(data1 != 0) > 0]
out1        <- iNEXT(data1.clean, q=c(0, 1, 2), datatype="abundance", endpoint = 100, nboot = 50)

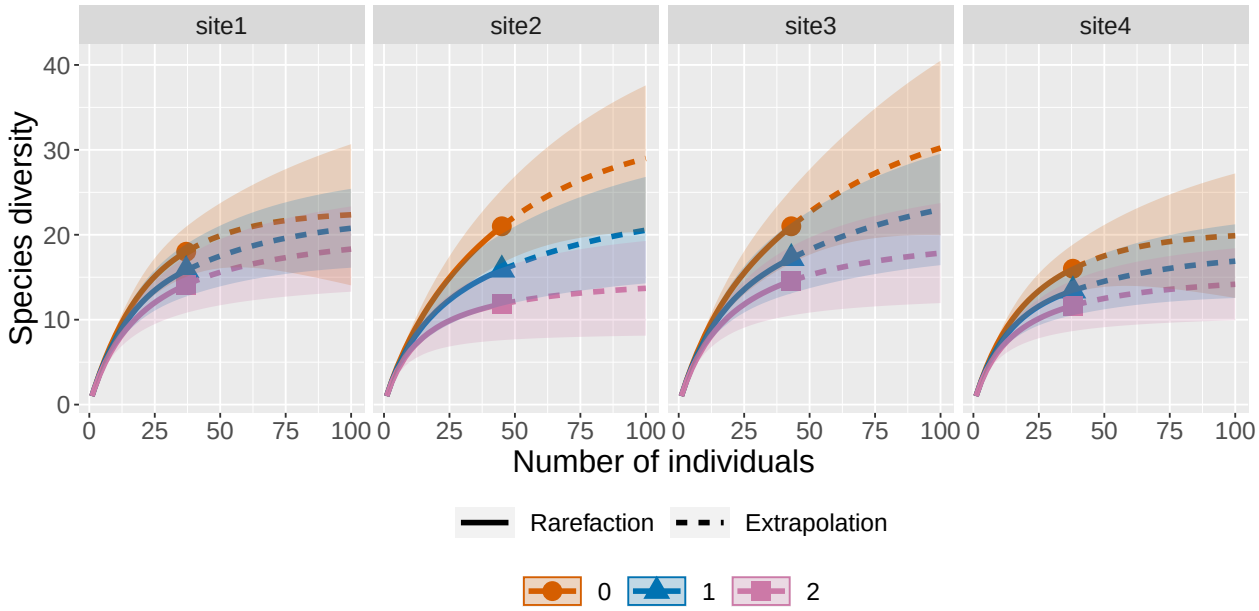
data2       <- t(comm_mat2$spec_dat)
data2.clean <- data2[, colSums(data2 != 0) > 0]
out2        <- iNEXT(data2.clean, q=c(0, 1, 2), datatype="abundance", endpoint = 100, nboot = 50)
```

Hill diversity

```
ggiNEXT(out1, type=1, color.var="Order.q", facet.var="Assemblage")
```

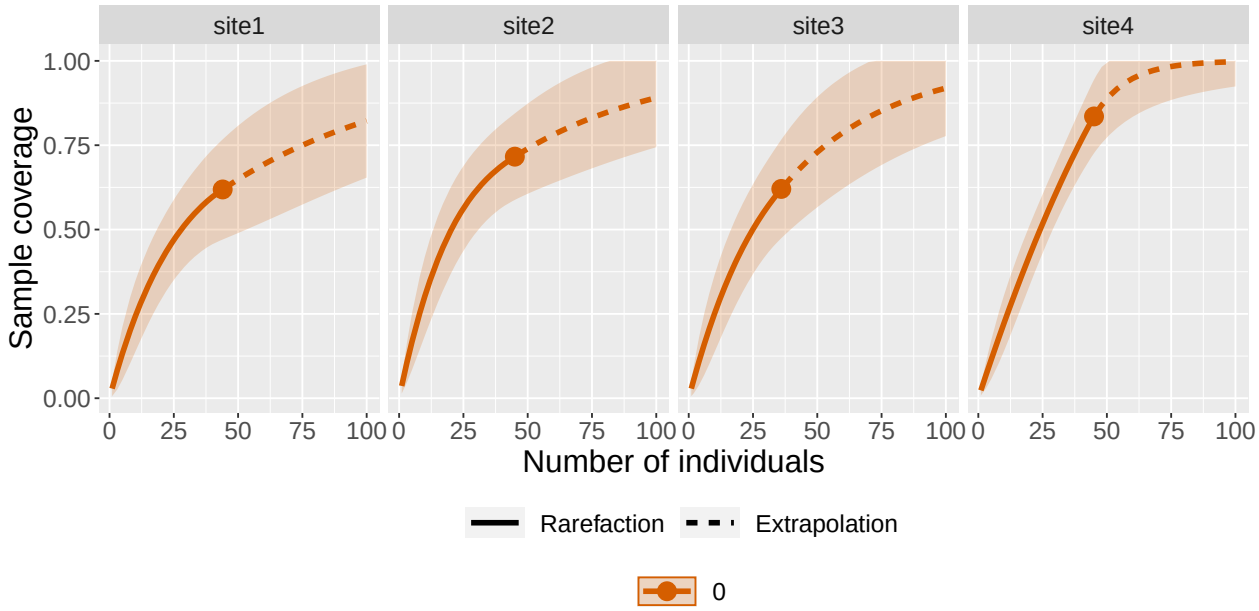



```
ggiNEXT(out2, type=1, color.var="Order.q", facet.var="Assemblage")
```

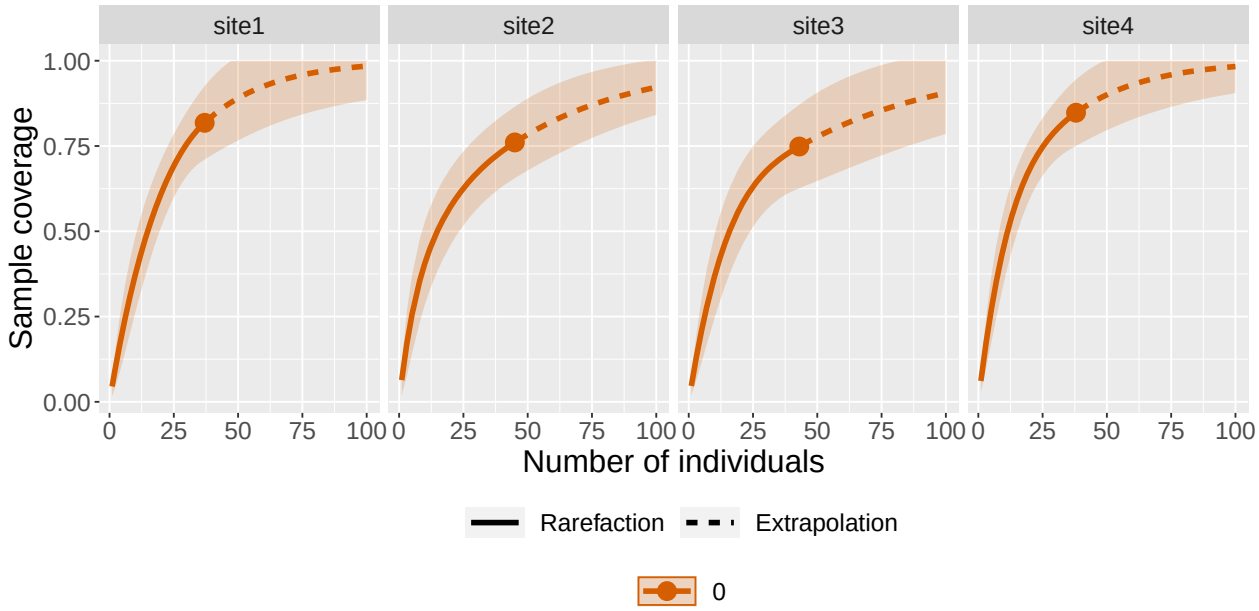


Number of individuals vs. coverage

```
ggiNEXT(out1, type = 2, color.var="Order.q", facet.var="Assemblage")
```

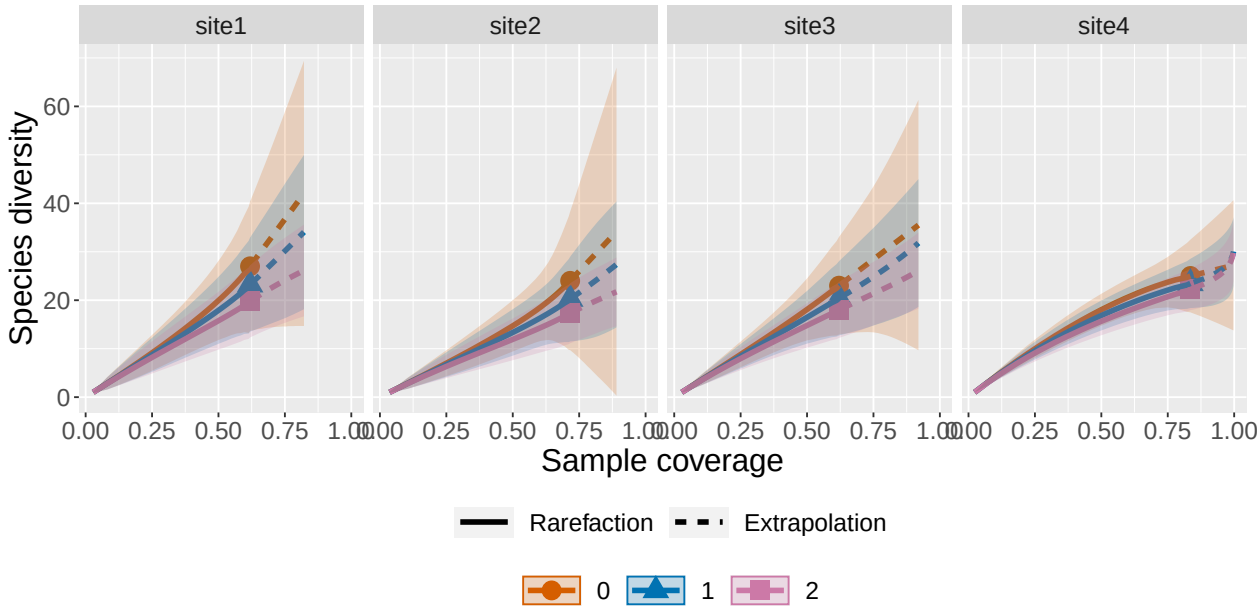



```
ggiNEXT(out2, type = 2, color.var="Order.q", facet.var="Assemblage")
```

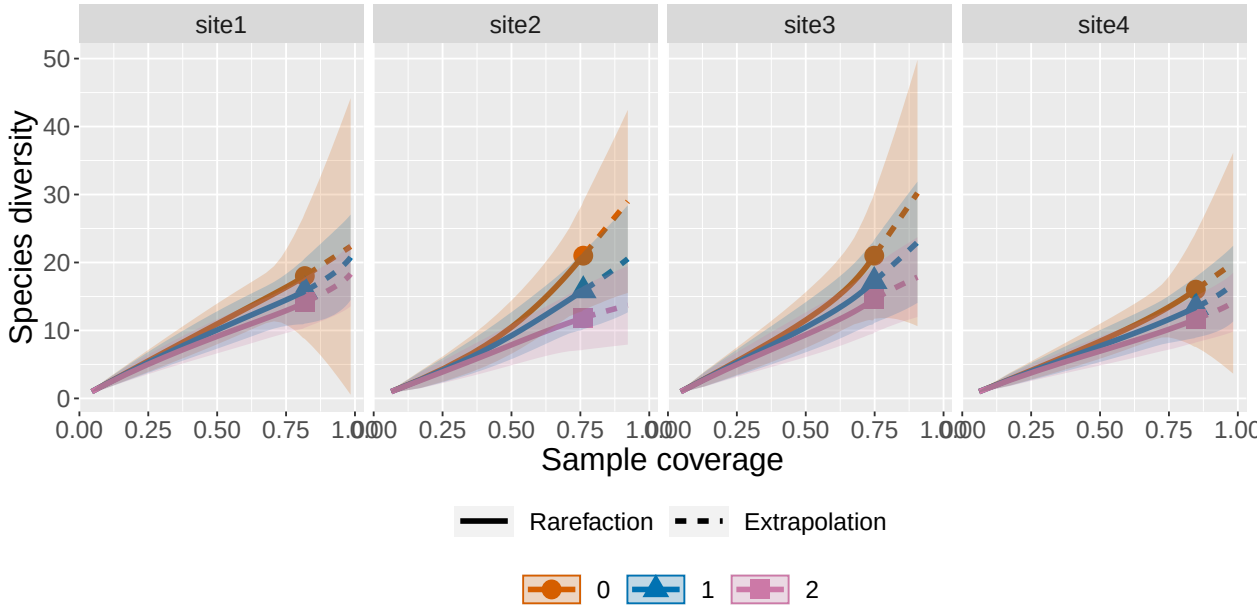


Standardization by coverage

```
ggiNEXT(out1, type = 3, color.var="Order.q", facet.var="Assemblage")
```




```
ggiNEXT(out2, type = 3, color.var="Order.q", facet.var="Assemblage")
```




```
# Print data to construct jelly communities
count1
```

```
##           count
## species1     14
## species10    11
## species11    22
## species12    17
## species13    19
## species14    15
## species15    19
## species16    22
## species17    13
## species18    21
```

```
## species19    14
## species2     18
## species20    14
## species21    13
## species22    12
## species23    15
## species24    18
## species25    14
## species26    17
## species27    14
## species28    13
## species29    22
## species3     14
## species30    21
## species4     23
## species5     17
## species6     24
## species7     18
## species8     12
## species9     14
```

```
count2
```

```
##           count
## species1     79
## species10    15
## species11    12
## species12    17
## species13    19
## species14    17
## species15    21
## species16    12
## species17    17
## species18    15
## species19     5
## species2     59
## species20    13
## species21    11
## species22     8
## species23     3
## species24     9
## species25     5
## species26     6
## species27     3
## species28     4
## species29     3
## species3     31
## species30     5
## species4     22
## species5     25
## species6     11
## species7     16
## species8     14
## species9     23
```

Beta diversity

```
combined_data <- comm_mat1$spec_dat %>%
  bind_rows(comm_mat2$spec_dat) %>%
  replace(is.na(.), 0)

com.structure   <- metaMDS(combined_data[,1:30], distance = "bray")
```

```
## Run 0 stress 0.08426349
## Run 1 stress 0.1009471
## Run 2 stress 0.1423353
## Run 3 stress 0.1416455
## Run 4 stress 0.08426349
## ... Procrustes: rmse 4.201763e-05  max resid 7.004889e-05
## ... Similar to previous best
## Run 5 stress 0.0882584
## Run 6 stress 0.0882584
## Run 7 stress 0.08426349
## ... New best solution
## ... Procrustes: rmse 2.207607e-05  max resid 3.774662e-05
## ... Similar to previous best
## Run 8 stress 0.08426349
## ... Procrustes: rmse 1.947131e-05  max resid 3.482173e-05
## ... Similar to previous best
## Run 9 stress 0.08426349
## ... New best solution
## ... Procrustes: rmse 1.566854e-05  max resid 2.768208e-05
## ... Similar to previous best
## Run 10 stress 0.1009472
## Run 11 stress 0.08426349
## ... Procrustes: rmse 1.850001e-05  max resid 3.187615e-05
## ... Similar to previous best
## Run 12 stress 0.0842635
## ... Procrustes: rmse 2.480587e-05  max resid 3.747313e-05
## ... Similar to previous best
## Run 13 stress 0.08825857
## Run 14 stress 0.08825851
## Run 15 stress 0.1423353
## Run 16 stress 0.08825841
## Run 17 stress 0.1009474
## Run 18 stress 0.08825841
## Run 19 stress 0.08825854
## Run 20 stress 0.08426349
## ... New best solution
## ... Procrustes: rmse 9.056922e-06  max resid 1.598586e-05
## ... Similar to previous best
## *** Best solution repeated 1 times
```

```
spp.scores      <- as.data.frame(scores(com.structure)$sites)

spp.scores.data <- spp.scores %>%
  mutate(community = rep(c("community1", "community2"), each = 4)) %>%
  mutate(sites = rep(c("site1", "site2", "site3", "site4"), times = 2))
```

```
ggplot(spp.scores.data, aes(x = NMDS1, y = NMDS2, shape = community, color = sites))+
  geom_point(size = 6)+
  theme_bw()
```

